

Prime Palindrome Program

Java Lab Record

1. Aim

To develop a Java program to check whether a given number is both a prime number and a palindrome using loops, conditions, and arithmetic operations.

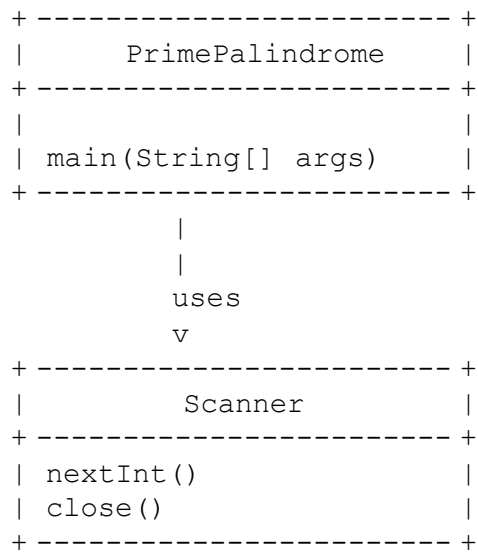
2. Step-by-Step Procedure

1. Import the Scanner class to accept input from the user.
2. Create the PrimePalindrome class containing the main() method.
3. Create a Scanner object.
4. Read a number from the user.
5. Initialize count to zero.
6. Use a for loop from 1 to the given number to count its divisors.
7. If the number is exactly divisible by the loop variable, increment count.
8. Store the original number in a temporary variable.
9. Reverse the number using a while loop by extracting each digit.
10. Compare the divisor count with 2 to check whether the number is prime.
11. Compare the original number with its reversed value to check whether it is a palindrome.
12. If both conditions are true, display Prime Palindrome.
13. Otherwise, display Not a Prime Palindrome.
14. Close the Scanner object.
15. Stop the program.

3. Algorithm

1. Start.
2. Import java.util.Scanner.
3. Create a Scanner object.
4. Read the number num.
5. Set count = 0.
6. For i = 1 to num, if num % i == 0, increment count.
7. Set temp = num and rev = 0.
8. While temp > 0, find the remainder using temp % 10.
9. Update rev = rev × 10 + remainder.
10. Update temp = temp / 10.
11. If count == 2 and num == rev, display Prime Palindrome.
12. Otherwise, display Not a Prime Palindrome.
13. Close the Scanner object.
14. Stop.

4. UML Class Diagram



Variables

```
used: num
```

```
: int
```

```
count :
```

```
int temp
```

```
: int
```

```
rev : int
```

```
rem : int
```

```
i : int
```

5. Code

```
import java.util.Scanner;

public class PrimePalindrome {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        // Create Scanner object for user input.

        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        // Read the number entered by the user.

        int count = 0;
        // Initialize divisor count to check for prime.

        for (int i = 1; i <= num; i++) {
            // Check all possible divisors of num.
            if (num % i == 0)
                // If num is exactly divisible, it has a divisor.
                count++;
            // Increase the divisor count.
        }

        int temp = num;
        // Keep the original number for comparison.
        int rev = 0;
        // Initialize the reversed number.

        while (temp > 0) {
            // Reverse the number digit by digit.
            int rem = temp % 10;
            // Extract the last digit.
            rev = rev * 10 + rem;
            // Add the digit to the reversed number.
            temp = temp / 10;
            // Remove the last digit from temp.
        }

        if (count == 2 && num == rev)
            // Check prime and palindrome conditions.
            System.out.println("Prime Palindrome");
        else
            System.out.println("Not a Prime Palindrome");

        sc.close();
        // Close the Scanner object.
    }
}
```

6. Sample Output

Sample execution using the input 131. The output is presented in the same console-style format as the reference.

```
--- Prime Palindrome Program ---
```

```
Enter a number: 131
```

```
Prime Palindrome
```